CABIOS

Vol 2 no 1 1986
Pages 23 – 27

A discrete model of bacterial metabolism

M R Watson

Abstract

This paper describes a computer model of the intermediary metabolism of bacteria during steady-state growth and during adaptations, e g. to new carbon sources. Metabolic regulation is represented as a process of optimisation, in which the trend is towards improved metabolic performance. The model uses linear programming techniques for the optimisation The implementation falls into four phases: (i) assembly of model parameters; (ii) calculations; (iii) storage of solutions and (iv) projection of solutions. The use of a commercial database and a commercial spreadsheet has proved to be of great assistance in the first and third phases. A metabolic map format, with the optional addition of conversion values, names of enzymes or co-factors has been used to project the results in a form convenient for inspection.

Introduction

A consideration of the molar yields of ATP from various pathways forms part of most courses and texts on metabolism, because it throws a considerable amount of light on the way in which catabolism provides the energy for vital processes It is less usual to try and form a complete balance for the generation and utilisation of ATP at the whole-cell level, although molar growth yields of bacteria have been measured (Bauchop and Elsden, 1960) and balances for a range of bacteria have been estimated (Forrest and Walker, 1971)

Given the complexity of metabolism, especially when biosynthetic routes are active, manual calculations of such balances can become very laborious It is possible to obtain support from the computer; indeed, by the use of the techniques described below it is possible to consider balances not just for ATP but for other metabolites (33 in the present case). These calculations provide students with a useful demonstration of the application of computing techniques in biochemistry

There are usually many different distributions of metabolic flux which can provide the required outputs. For example, in many organisms ATP may be generated by anaerobic glycolysis or by oxidative phosphorylation, depending on the environmental conditions. Closer inspection reveals that there are intermediate states, in which ATP generation occurs with some uptake of oxygen and some release of fermentation products (aerobic glycolysis). Metabolic adaptations, e.g from anaerobic to aerobic, may thus be modelled as a series of transitions through a discrete set of states. What initially seemed to be a balancing of inputs against outputs has broadened out into a modelling procedure.

A computer program of this type and its use in teaching has already been described (Watson, 1984a, 1984b) The relevant algorithm was not fully elaborated, and further details are given here Also it has become apparent that the user interface (display and editing of model parameters, and the collection, editing and projection of results) required much improvement, for the facilities provided in the previous software were rudimentary. The programs are now used in conjunction with commercial general-purpose data base and spreadsheet packages to provide better facilities for users. An account of these improvements is also provided.

System and Methods

The programs are written in Sinclair Superbasic with machine code extensions constructed using the Metacomco Assembler Development Kit for the QL, release 1.7 (Metacomco, Bristol, UK), using many of the firmware routines described by Dickens (1984). The application also requires the Archive database, version 2.0 and the spreadsheet, Abacus, version 2 00 both by Psion Ltd. A medium resolution colour monitor (Sanyo model CD 311MB) is used as VDU. Hard copies of the diagrams were obtained using a QL Centronics printer interface (Miracle Systems Ltd., Bristol, UK) designed to connect the serial 1 port of the computer to the parallel interface of an Epson FX-80 printer.

Algorithm

The models are designed to explore the range of alternative steady states through which bacterial cells can harness energy metabolism to growth under a given set of growth conditions. The computer is programmed to select one particular state as that most likely to represent the metabolism of the organism. This range of states is different in different organisms, because of inherited differences in the range of pathways present; even in a single species the range of states is continually expanding or contracting, as cellular controls open up new pathways or close down existing ones Thus there is a continuous search

Department of Biology and Biochemistry, North East London Polytechnic, London E15 4LZ, UK

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for new optimal states, and we seek to represent metabolism and its changes in these terms. A suitable mathematical framework for this type of modelling is provided by linear programming Dantzig (1963) has described the main steps in formulating a linear programming model.

Firstly, the system is broken up conceptually into a number of elementary functions that are called 'activities' For metabolism, it is convenient to equate these activities with the net chemical changes effected by single enzyme catalysts, although it is also convenient to equate some activities with the operation of whole pathways, to reduce the total number of activities considered to a manageable level. This process of assignment therefore involved simplification. This is always necessary in linear programming, the rule is to try to separate primary and secondary features, simplifying the secondary features first For example, in this model of energy metabolism, it seemed appropriate to delineate the reactions in the central area of metabolism (glycolysis, the tricarboxylic acid cycle) clearly as individual activities, merging many other reactions into pathway activities. There is too much detail to allow a full account of all the assumptions made, but the database (see below) can be inspected and modified if necessary, as the question of what constitutes a proper simplification is subject to individual judgement and experience.

Secondly, the model must describe the inputs and outputs required from the pathways under the given growth conditions, i.e a statement of constraints is necessary

Thirdly, the model must contain a performance criterion, called an objective function, to guide the model in its search for the optimal state. A criterion of minimal free energy loss has been used in the models. It is doubtful whether such a simple criterion (thermodynamic efficiency) is wholly realistic, for the factors which have shaped the control characteristics of pathways probably reflect other requirements for effective operation, but it generates models which are useful for teaching purposes

When these three steps have been completed the problem is encoded in the form stated in equation (i) ($i = 1,2,\ldots,m$, r = activities, m = metabolites.)

$$\sum_{j=1}^{r} a_{i,j} \quad x_j = b_i \qquad \text{(i)}$$

The nature and magnitudes of the activities present are contained in the $x_{(j)}$ terms on the left. The magnitudes are not rates, but net conversions by the relevant enzyme/pathway over a period of steady-state growth [these quantities have previously been encountered in irreversible thermodynamics, where they are known as 'advancements' (Prigogine, 1967)] All the information relating to the relative proportion of reactants/products, the stoichiometry, is contained in the $a_{(i,j)}$ matrix. Each of the elements of this matrix (Heinrich *et al.*, 1977) is a stoichiometric coefficient, each column containing the elements relating to one activity. The constraints are represented by the $b_{(i)}$ elements. These are calculated at an early stage from compositional information on the organisms, and information on molar growth yields (Bauchop and Elsden, 1960) using a spreadsheet (see below). The objective function is represented by the inequality

$$\sum_{j=1}^{r} c_j \quad x_j > z\ (\min) \qquad \text{(ii)}$$

in which the $c_{(j)}$ coefficient is a free energy cost associated with a unit value of the activity j, and $z(\min)$ is lower bound to the free energy loss, which the model is continually seeking to approach

The calculation to be carried out can be summarised as follows. Given the coefficients in the above equations, and information on the range of activities available to the optimising process, which activities contribute to the optimal state, and what are the associated conversion values $x_{(j)}$?

The calculation is carried out using the revised simplex method, details of which may be obtained from Garvin (1960). It requires the selection of a trial set of activities at the start. If this selection is not optimal over the range of activities present, progressive substitution of one activity for another takes place until the optimal state is arrived at

Implementation

The implementation takes place in four stages, each of which will now be described.

Assembly of model parameters

The values for the coefficients for the $a_{(i,j)}$ matrix and the objective function are stored as a total of 75 records in the database Each record contains the coefficients for one activity — see Figure 1 for an example. Assembly of the matrix takes place automatically, under the control of a procedure written in the database language It takes 6–8 min in all Reconstitution of the matrix is not required for every run, normally facilities for the addition of activities at the next stage is sufficient to represent the metabolism of different organisms.

The amounts of intermediates required for bacterial growth are also stored in the same database file, but their calculation from compositional data on bacteria and from molar growth yields by manual methods is excessively laborious. The calculations may, however, be carried out rapidly and conveniently using the spreadsheet, once suitable files have been prepared The database and spreadsheet files are all encoded on a single microdrive cartridge

Calculations

A statement of the range of activities to serve as a starting point for the calculations is made using a program written in SuperBasic, recorded on a second microdrive cartridge. Menus pro-

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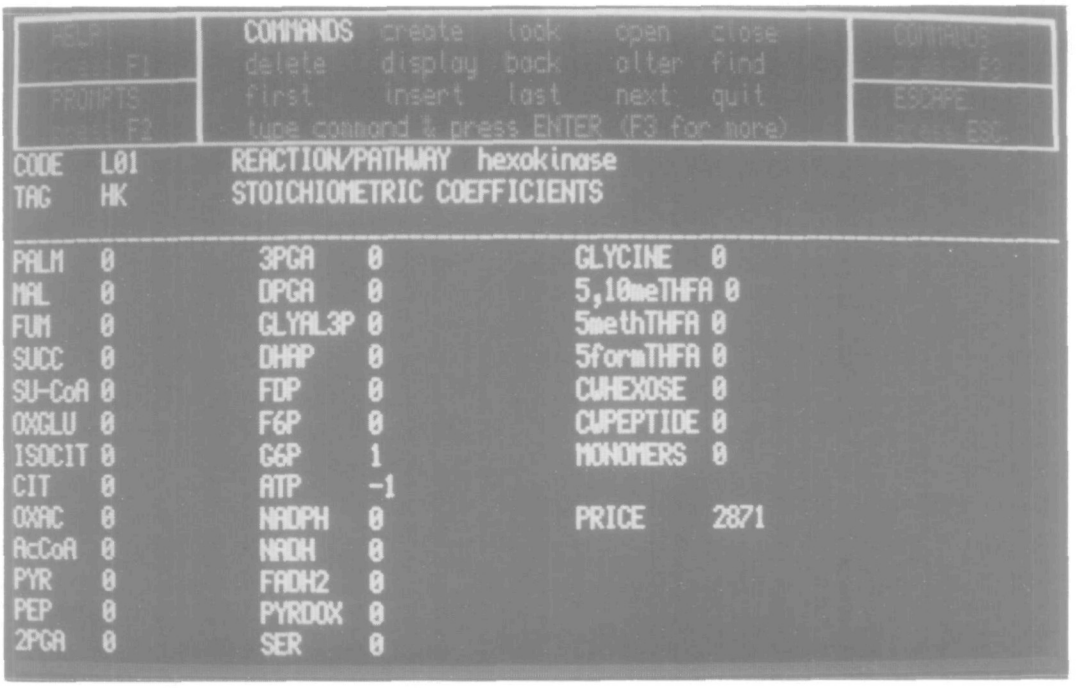

Fig. 1. Database record describing the stoichiometry of the reaction catalysed by hexokinase Letters indicate identities of the metabolite concentrations kept in balance in the models The value entered against price is the relevant coefficient for the objective function

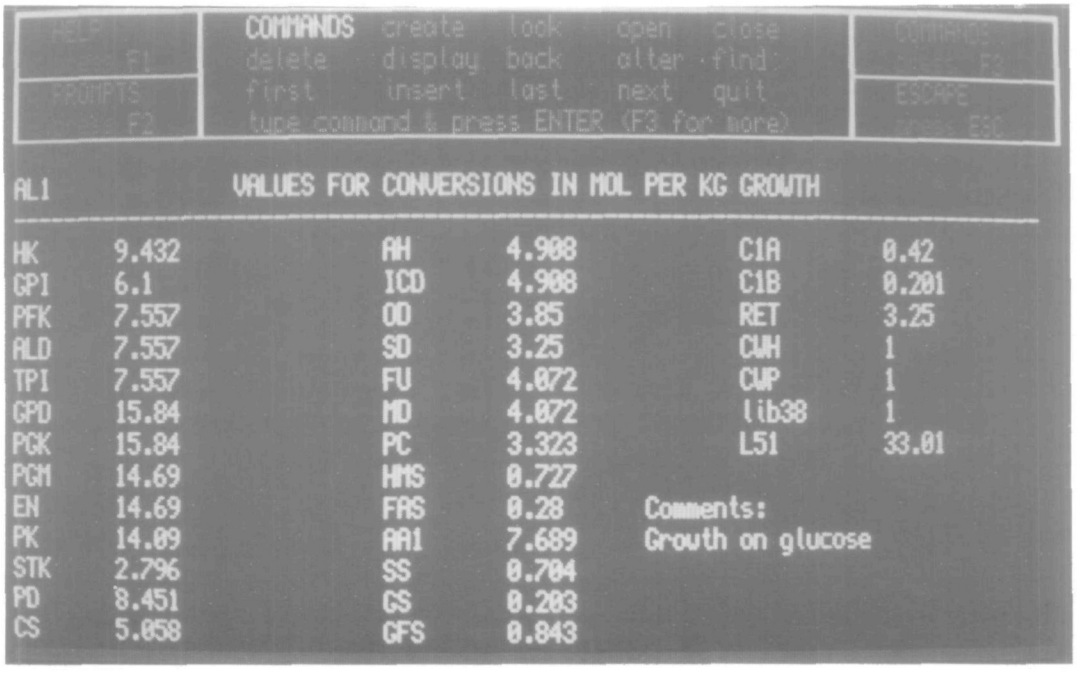

Fig. 2. Database record describing the conversions in the central pathways for growth on glucose as sole carbon source Letters stand for enzymes catalysing the relevant conversion, numbers are conversion values (see text)

vide convenient facilities for the addition of specialised activities, i.e those with limited biological distribution. The simplex substitution cycle take ~2 min and the progress of substitutions is indicated on the VDU screen All the metabolic states with their associated conversion values are recorded automatically on the microdrive cartridge on optimisation.

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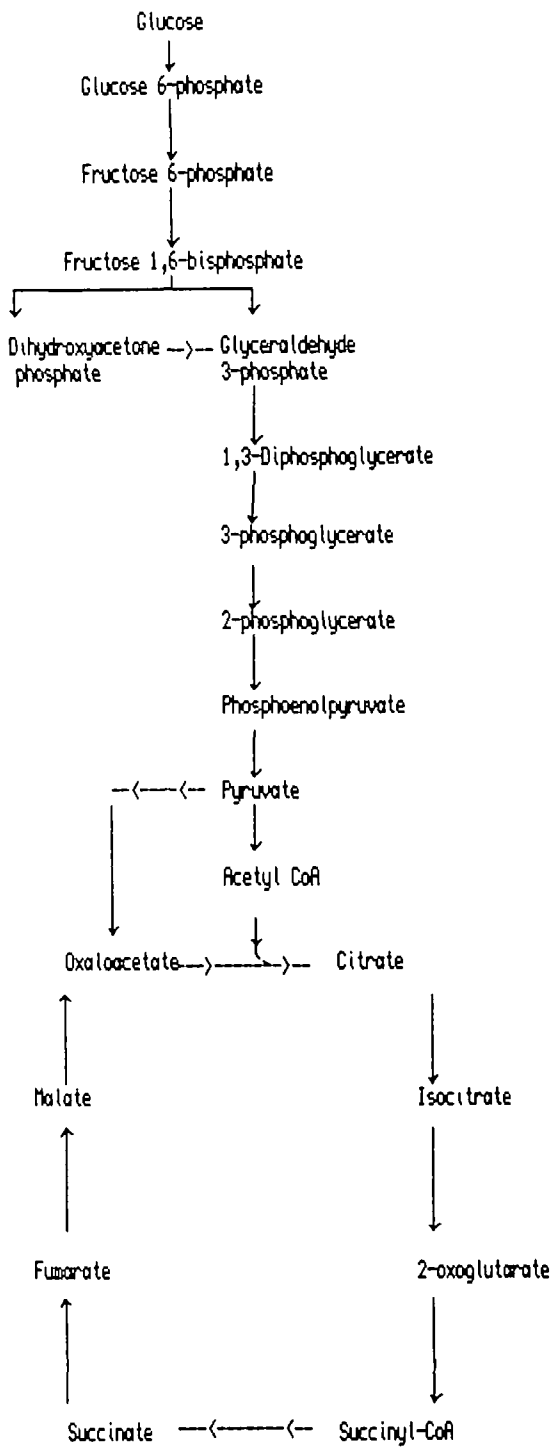

Fig. 3. Diagram projected on the VDU screen by graphics program in response to input of the data shown in Figure 2 The diagram has been printed in three vertical sections and the hard copy has been reduced by photocopying

Storage of solutions

The results may be inspected using the standard facilities of the database, using a screen file to project the values in a convenient format (Figure 2) This allows convenient inspection of feasibility (absence of negative values for activities), rejection of unwanted solutions and the addition of titles to aid in the identification of the work at a future date

Projection of solutions

The biochemical significance of the solutions obtained can normally be seen more readily from a metabolic pathway diagram than from the numerical estimates. The method used to produce the output is to design pathway diagrams in the form of a mosaic, in which the printing of each element in the design is governed by the value of a single conversion value Only non-zero positive conversions will trigger the printing of the corresponding map element. In this way a large number of maps can be generated from a modest number of elements.

The diagrams produced using the earlier version of the programs (Watson, 1984a, 1984b) were designed to fit into the high-resolution VDU display of the Apple II. A highly condensed notation of rectangles for individual reactions, with only a few letters to indicate the names of intermediates, was employed. In order to obtain a more intelligible output, maps occupying more than the area of a single screen are now used The display may be scrolled so that it may be viewed or printed out in sections. Names of enzymes are also printed in colour-coded areas, a single key depression leads to the appearance of names of co-factors instead of enzyme names, or numerical values for the magnitudes of conversions. Maps lacking these coloured additions may also be obtained (Figure 3).

Discussion

Comparison with kinetic models

The present models are of the discrete type (Turing, 1948), representing metabolic change as a discontinuous process of transition between different metabolic states Most existing metabolic models are of the continuous (kinetic) type, representing metabolic change as a smooth progression in reaction rates and metabolite concentrations. Kinetic models of metabolism have been reviewed by Heinrich *et al.* (1977) and by Wright and Kelly (1981).

We can think of the digital computer primarily as an extraordinarily rapid and effective tool for computation. In that case it is natural to see in it a means of extending familiar calculations from chemical and enzyme kinetics to multi-enzyme systems This has been the general trend. Alternatively we may see in it a totally novel type of mechanism, suggesting new ways of representing the living process. In that case we try to build, as at present, new types of biochemical models There is also a difference in the computation methods. The continuous (kinetic) models are based on the dynamic laws for chemical reactions, as embodied in the principles of chemical and enzyme kinetics. In the discrete model transitions between configurations are governed by an objective function. The metabolic controls are supposed to have evolved in such a way that they tend

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to conserve chemical energy This is in line with current biochemical thinking (Atkinson, 1971, 1977), i e. cellular metabolism is like an economy, in which chemical resources are automatically disposed to the best effect.

There is a pressing need to make full use of the emerging computer technology in biochemical courses. The continuous models are not very suitable for this, because they run very slowly on all but the most powerful systems. The discrete model presents us with a practical approach using a small computer system. The discrete model, while convenient for teaching purposes, cannot substitute for kinetic models. The new physical techniques for monitoring reaction rates and metabolite concentrations in the living state (Gadian, 1983) are now providing high, quality data which can be expected to erode the diffuseness surrounding our understanding of enzyme kinetics *in vivo*. This work requires support from kinetic theory, for the discrete models do not specify metabolite concentrations, nor do they provide a time-scale for events. We can expect much progress with the kinetic models as speculation about *in vivo* kinetics hardens into concrete knowledge.

The present model is not intended to represent the metabolism of specific microbes in detail To adapt it for this purpose requires a detailed study of every facet of the metabolism of the organisms in question, and a great deal of patience, for it is known from experience with economic applications that the construction of linear programming models of specific processes is a tedious and exacting task (Orchard-Hays, 1968). For more realistic discrete models, it may be best to study the metabolism of mammalian organs, where the greatest volume of high-quality data is accumulating, and where the practical needs for a detailed understanding are greatest

Software

Experience gained during this work has indicated the magnitude of the programming effort required to support a teaching model of metabolism. The design of software for teaching purposes is particularly demanding, for waiting times of even a minute or two can seem interminable in a classroom situation, and facilities for the projection of results must be clear and unambiguous. Taking into account the supporting commercial packages, and the database and spreadsheet files used in conjunction with them, the total size of the present package in bytes is about four times that of the original Apple II version. This has resulted in an improved user interface, but the level of detail in the models is unchanged. As the package has grown, the use of machine code subroutines to speed up slow sections of the procedures, and to give more informative displays has become more and more essential.

The use of general purpose commercial sofware packages can ease the labour of producing suitable packages. The usefulness of spreadsheets for biochemical calculations was noticed by Kibby (1985). We are still left, however, with writing programs to interface the commercial packages with routines carrying out special functions, such as the production of the map diagrams described above, and the linear programming. It is important when handling very large amounts of data to make as many of the transfers between the various sections automatic, both for rapidity and to avoid transcription errors. Packages which can operate in this way (integrated software) are becoming increasingly common, and this is an important development for specialised applications as well as for standard applications

Acknowledgement

I wish to thank Mr N Weedon of the Physical Sciences Department of the Polytechnic for the photographs of the VDU screen

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Received on 22 August 1985, accepted on 3 January 1985

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